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Future changes in rice yield over Kerala using climate change scenario from high resolution global climate model projection

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The impact of climate change on agricultural yield is one amongst the major concerns the world is witnessing. Our study focusses on rice yield prediction for an agricultural research station in Kerala with the help of climate change scenario input from the Meteorological Research Institute (MRI) Global Climate Model (GCM) projection under Representative Concentration Pathway 8.5 (RCP8.5). We have used Cropping System Model (CSM) Crop Estimation through Resource and Environment Synthesis (CERES) Rice within Decision Support System for Agrotechnology Transfer (DSSAT) package for predicting the yield. Our study has the novelty of using very high-resolution climate data from a model which is highly skilful in capturing the present-day climate features and climatic trends over India (in particular, over the Western Ghats), as input for simulating the future crop yield. From this study, we find that the rice yield decreases due to rise in temperature and reduction in rainfall, thereby reducing the crops maturity time in the future. Based on our results, the adaptation measures suggested to achieve better yield under future warming conditions are: (i) to opt for alternative rice varieties which have tolerance to high temperatures and consume less water, and (ii) shifting of planting date to the most appropriate window.

Keywords. Global Climate Model; climate change; cropping system model; rice yield; adaptation measures.

1. Introduction

Study on climate alone would appear surreal without realising the value of science in climate simulations for which a quantitative assessment of climate's impact is required (Gadgil and Gadgil 2006). A change in the natural climate is due to change in earth's revolution, variation in solar insolation and other natural factors. But over the last few years, a dramatic change in daily weather has been noted. This is deemed to be by human influence, where greenhouse gases are emitted into the atmosphere at an alarming rate by burning forests, grasslands and fossil fuels. The effect of climate change invariably affects the agricultural yield due to change in climate variables which are most responsible for the growth and development of crop.

The population of the world is expected to rise to 8.3 billion by 2030 and 9.1 billion by 2050. Global

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food productions should go in line with the increase in demand by about 40% by 2030 and 70% by 2050 (FAO 2009). With respect to India, the population is expected to go up to 1.6 billion by 2050, therefore there needs to be increase in the productivity, especially in rainfed areas of India (Aggarwal *et al.* 2008). In an agrarian economy like India, the gross domestic product (GDP) is greatly dependent on timely monsoon (Gadgil and Gadgil 2006), and other climatic parameters like temperature and solar radiation. Although the contribution of agriculture to the GDP has reduced from over 50% post-independence to 22% in 2000, the impact it has on GDP remains the same and therefore needs to be taken care (Gadgil and Gadgil 2006). Therefore, the impact study of change in climate parameters on crop yield is inevitable.

The challenges in food and environment safety can only be envisaged with smart and adaptive agricultural practices. The major constraints for agricultural productions are low soil fertility, non-uniform distribution of rainfall, droughts, etc., apart from the current increasing trend of greenhouse gas emission. Thus, the incorporation of knowledge about future climate change in agricultural sector is necessary for better agricultural management. An increase in mean and extremes of temperature and rainfall are expected in major parts of the world (IPCC 2013). These changes can impact agriculture and food grain production of developing countries in tropics and subtropics (Parry *et al.* 2004; Mall *et al.* 2006).

Impact studies centred around climate change pose several questions as how accurate are the predictions given and the uncertainties associated with the prediction in both climate simulation and crop simulation. Global climate models (GCMs) in general predict a warmer and moister climate for the future with large uncertainties. Study by Battisti and Naylor (2009) confirms that decline in food production is due to seasonal heat that are extreme, which may become perennial in the long run if no adaptive measures are followed. A more integrated outlook into climate and crop modelling in a probabilistic framework is gaining its relevance in changing climate.

Rice is regarded as the major food consumed by more than half of the world, especially in Asia (Schiller *et al.* 2006). The major parameters responsible for rice production are temperature, rainfall, CO_2 and solar radiation (Fageria 2007). Razzaque *et al.* (2011) showed that, the yield in rice increases with increased CO_2 concentration and fixed temperature. This positive effect was nullified by the rise in temperature due to greenhouse effect. Nigam and Mishra (2003) showed the correlation of weather parameters and rice yield over Pantnagar region and found that minimum temperature, maximum temperature, minimum humidity and rainfall have positive correlation with rice yield.

Questions on how the future climate is going to be, how the warmer environment is going to affect the moisture transport over the tropics, how the rainfall is going to be distributed over the annual cycle, etc., will have greater role in gauging future impacts of climate change on rice crop. High temperatures are going to challenge the demand for food with increase in population, if adaptation measures are not taken. Many crop modelling studies (Semenov and Porter 1995; Easterling *et al.* 1997; Semenov and Barrow 1997; Porter and Semenov 2005) have shown the importance of recognising climate change as an inevitable factor in climate impact studies.

The Coupled Model Intercomparison Project (CMIP) Assessment Report 5 (AR5) models project an increase of 3–4°C temperature over central Asia by the end of 21st century along with an increase in CO_2 concentration up to 485–1000 ppm. With an increase in CO_2 concentration, the yield and biomass of rice is expected to increase (Kim *et al.* 2003; Razzaque *et al.* 2011). Parry *et al.* (2007) study shows that under such circumstances, the crop yield decreases by 30% even after considering the direct physiological effect due to increase in CO_2 concentration. Krishnan *et al.* (2007) showed that for 1°C of increase in temperature, the rice yield declines by 7.2% with the current trend of CO_2 concentration but with increase in CO_2 up to 700 ppm, an average increase in yield by 31% was found over India.

It is important to achieve higher yield to meet the requirement of the growing population. Higher yield can be achieved by increasing the total crop biomass. The total biomass on the other hand is determined by crop photosynthesis and respiration loss, where both are sensitive to temperature. Thus, it is important to address the impact of higher temperatures induced by global warming on rice production. The effect of climate change on crop yields were extensively done using crop simulation models (Rosenzweig and Parry 1994). Scenario analysis of rice yield was done by Rajwade *et al.* (2015) for base period (1969–1999) and future periods 2020, 2050 and 2080 under A2 and B2

scenarios of HadCM3 GCM. The yield was found to decrease from base to future climate scenarios. Similar study was carried out by Satapathy *et al.* (2014) which also simulated the possible climate change impacts using HadCM3 data and found a continuous decline in the rice grain yield for the years 2020, 2050 and 2080 with increasing temperature >0.8°C over Kharagpur region. Data from HadCM3, CCCMA-CGCM2 and CSIRO Mk2 were used by Jalota *et al.* (2012) for yield prediction of rice over Punjab for mid-century and end of the 21st century under A2 scenario. They found that yield was much lower than average yield of base period and suggested for a shift in transplanting date to improve yields. Swain and Thomas (2015) reported a decrease in the yield of rice under various scenarios at different locations in India. Temperature stress was found to have greater influence on anthesis day in climate projection, whereas it is not so in baseline simulation. Challinor *et al.* (2007) study showed that significant correlation exists between duration of the crop and corresponding yield simulated. With the rise in temperature, an overall decline in rate of photosynthesis resulted in limited biomass accumulation (Squire 1990). It is to be noted that all these studies were carried out using climate inputs from coarse resolution GCMs which tend to hold large bias in simulating the present-day climate of South Asia, especially India. Coarse resolution GCMs have drawbacks in projecting the regional and seasonal rainfall distribution over India (Allen and Ingram 2002; Chou and Neelin 2004; Rajendran *et al.* 2013) which limits the representation of high-resolution interaction processes among circulation, local complex terrain, heterogeneous land surface and adjacent warm ocean. On the other hand, regional climate models (RCMs) of high resolution do not represent regional-global scale interaction due to lack of their feedback to the parent GCM and are known to depend on the lateral boundary forcing from the parent GCM. Therefore, a high resolution GCM is preferred as the alternative in such a situation.

Even after using improved varieties of rice for cultivation, the yield seems to be lower than the potential yield due to different levels of management practices and variation in climate conditions. Field experiments for identifying the impact in climate change on crop cultivation will take longer period of time. Thus, we resorted to crop models for this purpose. Station based weather data is regarded as the major source of data for crop simulation models (Sultan *et al.* 2005), but when needed for future climate scenarios, we opted for GCM output. We made use of high-resolution climate data from Meteorological Research Institute (MRI) GCM which is found to have high reliability in simulating present day climate of India (Rajendran *et al.* 2013, 2016) particularly the climate trends observed over the Western Ghats (WG) and Kerala, the southern-most state of India (Rajendran *et al.* 2012).

Amongst several crop simulation models available for impact studies, Decision Support System for Agrotechnology Transfer (DSSAT) (Jones *et al.* 2003) has proved to be versatile in having packages for all relevant crop varieties. Crop Estimation through Resource and Environment Synthesis (CERES) Rice module within DSSAT is used for rice crop estimation. There are several climate change impact studies using DSSAT for rice production in India (Subash and Mohan 2012; Mishra *et al.* 2013), but they lack sufficiently high-resolution climate inputs. This insufficiency suggests for alternative look out on the impact of climate change on rice cultivation using high resolution data from GCM, which has high reliability in simulating present day climate and observed climate trends over India. It also envisages the need to look for adaptive strategies for future to sustain maximum yield. Also, many of the studies reported climate change impact on rice production using CERES-Rice model (Swain and Thomas 2015) lacks a well validated, location-specific crop management for impact assessment. Through this study, we attempt to provide location specific cultivar management for climate change adaptation.

Section 2 deals with details of study area, climate data and soil data of the station, crop type and crop environment and crop simulation model used. Section 3 describes the effect of climate scenario on rice crop yield for the station considered as well as the major observations made and results obtained. Section 4 gives the conclusion of the study.

2. Models, datasets and methodology

In the present study, we have focussed on the most important crop of a particular region and its simulation using a crop model. Our major objective is to see how reliable climate data will help in providing a robust yield prediction which at

present is a major concern in reducing uncertainty associated with yield prediction.

2.1 *Study region*

A single model grid cell is taken up for this study. This enables in avoiding the perturbations caused by spatial variability and would only amount to climate or crop model perturbations. Field experiment data were collected from Agricultural Research Station (ARS) Mannuthy, Kerala, located at 10°32′N and 76°20′E at an altitude of 22 m above the sea level (figure 1).

2.2 *Climate of Vellanikkara*

This area comes under tropical climatic condition and is under the influence of southwest and northeast monsoon. This area receives a maximum rainfall during June to August (figure 2). The field experiment was done during *Kharif* season from May 2017 to November 2017 and the same data has been used for the current study. When crop models are applied to assess the impact of climate variability on crop growth, climate datasets are central for the same. For the location of interest, nearest weather station is used as a proxy. While using simulated data for climate change projection, depending on the model resolution, the location is fixed and the weather data under different climate regimes is derived such as daily precipitation, solar radiation, minimum temperature and maximum temperature.

Radiation data is required to determine the demand for evapotranspiration. Input for uppermost soil layer is rainfall. Minimum and maximum temperatures are used to find the daily mean temperature. Present-day climate data for data validation was taken from Principal Agro-meteorological observatory, Vellanikkara, Department of Agricultural Meteorology, College of Horticulture, Kerala Agricultural University. This long period climatology is plotted in figure 2. It shows daily climatological average of maximum temperature, minimum temperature, rainfall and solar radiation for the period 1994–2003.

2.3 *Soil data*

This is the second most important factor in crop model after climate data. Soil plays a vital role in determining water storage capacity. Soil is regarded as sandy clay loam type and comes under soil series laterite. The classification is given in table 1.

In order to see the chemical properties, soil samples were taken at two soil depths, 15 and 30

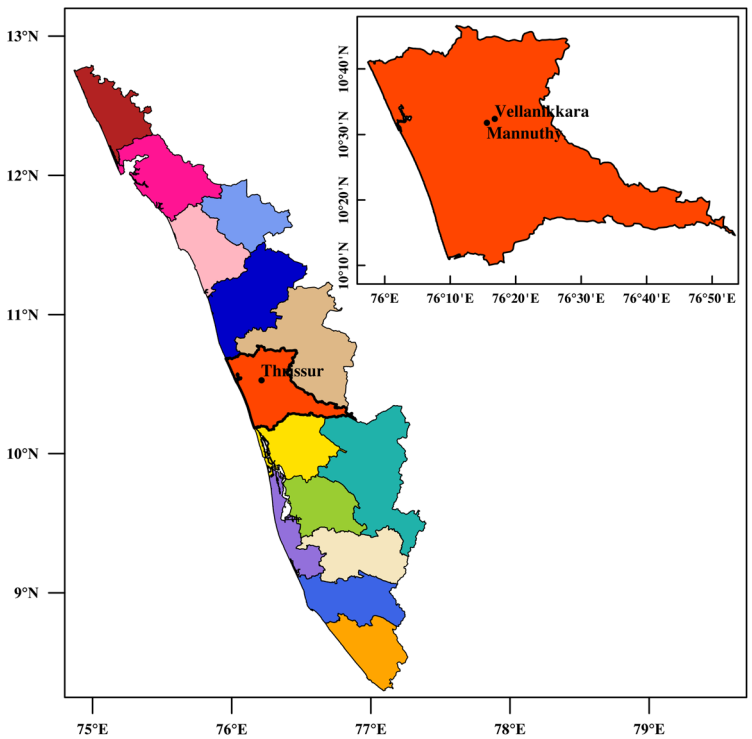


Figure 1. Inset: The study area of Agricultural Research Station (ARS) at Mannuthy and weather station at Vellanikkara within Thrissur district of Kerala. The state of Kerala with districts.

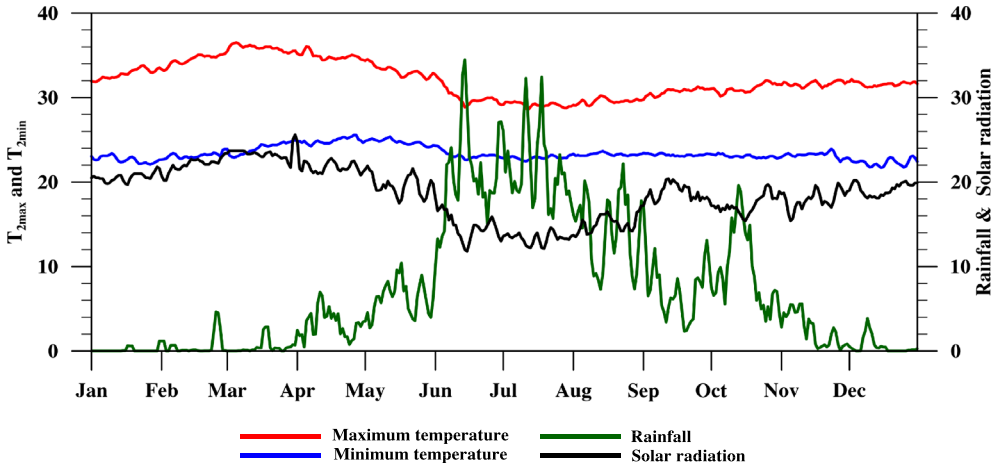


Figure 2. Climate variables at Vellanikkara station. Climatological (1994–2003) mean annual variation of rainfall (mm/day), solar radiation (MJm^{-2}/day), minimum and maximum temperatures (°C).

Table 1. *Physical properties of soil at the experiment site and their respective values in percentage (%).*

Sl. no.	Particulars	Value (%)
1	Coarse sand	27.6
2	Fine sand	24.3
3	Silt	22.2
4	Clay	26.0

Table 2. *Chemical properties of soil at the experiment site.*

Sl. no.	Parameter	Sample at 0–15 cm	Sample at 15–30 cm
1	pH	4.24	4.33
2	Organic carbon (%)	0.8	0.82
3	Available phosphorus (kg/ha)	116.81	113.29
4	Available potassium (kg/ha)	291.42	284.26

cm and was analysed for pH, organic carbon, phosphorus and potassium. The details are given in table 2.

2.4 *Crop type and crop environment*

Rice is a common crop that is frequently cultivated in the present study area. The experimental plot falls under the land type of *Kole* with cultivation during *Virippu* season (April–May to September–October). It generally has extra short duration, short duration and medium duration rice varieties. The current study focusses on short duration variety (110–125 days) *Jyothi* as it is one of the most popular rice varieties among farmers of Kerala. *Jyothi* is grown in all growing seasons because of its adaptability to various field conditions. It is a cross between short duration PTB 10 and high yielding genotype IR8. The environment chosen for a particular crop can vary with space and time. Hence, assumption of a homogeneous environment in terms of soil, climate and crop management would necessitate impact study of climate change on rice yield. The effect of temporal variability of climate needs to be addressed from such impact assessment studies. The field was laid out under split plot design with five dates of planting at 15 days interval, starting from 5th June to 5th August (D1–D5). Planting was done at a spacing of 15 × 10 cm. Transplanting to field area was done after 18 days.

2.5 *Phenophases of rice crop*

Phenology refers to the study of relationship between the dates of first occurrence of biological event in the annual cycle of crops with seasonal climatic changes. Mainly the phenophase of rice is divided into six stages based on morphological characters, such as:

- P1: Transplanting to active tillering,
- P2: Active tillering to panicle initiation,
- P3: Panicle initiation to booting,
- P4: Booting to heading,
- P5: Heading to 50% flowering,
- P6: 50% flowering to physiological maturity,

where P1 and P2 come under vegetative period, P3–P5 come under reproductive period and P6 comes under ripening period. Table 3 shows the

Table 3. *Phenophases of rice crop variety Jyothi. Duration of different phenophases during crop growth and development.*

Phenophases	Days after transplanting
Active tillering	27
Panicle initiation	36
Booting	57
Heading	63
50% flowering	67
Physiological maturity	103

duration of different phenophases for rice variety *Jyothi* as was observed in the experiment.

2.6 *Climate model, RCP8.5 projection and data*

Data from MRI-GCM (Mizuta *et al.* 2012) simulation which is an Atmospheric Model Intercomparison Project (AMIP) type simulation, with ∼20 km horizontal resolution corresponding to triangular truncation of T_L959 (1920 × 960) with linear Gaussian grid in the horizontal, is used in this study. For the present-day simulation, the model is forced with observed historical sea surface temperature (SST) and sea ice concentration (1979–2009) from Hadley Centre data set version 1 (HadISST1) (Rayner *et al.* 2003). Future projection was made using time slice experiment with forced external boundary conditions (SST from CMIP5 models). The boundary SST data was constructed as in Mizuta *et al.* (2014). The model was integrated under Representative Concentration Pathway (RCP) 8.5 scenario (Van Vuuren *et al.* 2011; Taylor *et al.* 2012) at 20 km resolution. The datasets used as input for crop simulation are daily total precipitation, maximum surface air temperature at 2 m, minimum surface air temperature at 2 m, and solar radiation. The climate change simulation by MRI-GCM produces an overall intensification of summer monsoon rainfall over India, but a reduction in rainfall over the WG region which is near to the area of present study (Rajendran and Kitoh 2008; Rajendran *et al.* 2012, 2013, 2016). This is consistent with the works that show rainfall during summer monsoon season increases over most parts of India at the same time with a decreasing trend over WG region with 90% significance (Guhathakurta and Rajeevan 2008; Rajendran *et al.* 2012). Many of the current RCMs (e.g., CORDEX-SA model) fail to capture this decreasing trend. Besides, the skill of the high resolution GCM in simulating the present-day climate also puts confidence in using its output rather than the output of an RCM. Singh *et al.* (2017) show that when compared to parent GCM, the CORDEX RCMs do not have any value addition due to downscaling. The mean Indian summer monsoon rainfall trend is not captured by RCMs when compared to their parent GCMs, especially over WG. Moreover, CORDEX RCMs lack very high resolution (only ∼40 km), but our GCM is having 20 km horizontal resolution. As a result, MRI 20 km can be considered superior to other state-of-the-art climate models. The crop yield was simulated for present day (1994–2003), and end of the 21st century (2090–2099).

As in Varghese *et al.* (2020), figure 3 shows the Taylor diagram (Taylor 2001) that compares the skill of 20 km model with Yoshimura parameterization scheme (YS) (Yoshimura *et al.* 2015) which is found to be superior than other state-of-the-art models participated in IPCC AR5. The study by Varghese *et al.* (2020) demonstrated the skill of 20 km MRI model in simulating the monsoon rainfall with the least bias and highest pattern correlation coefficient against the observation, when compared with other state-of-the-art global climate models such as CMIP5 models and MRI 60 km models. It can be seen that it is closer to the observation compared to MRI 60 km model simulations as well. MRI model at 20 km is thus having the best skill owing to its ultra-high resolution and better representation of data over the station under study.

Validation of MRI data with 1994–2003 climatology of Vellanikkara on various meteorological parameters is carried out. Figure 4 shows climatological annual cycle of maximum temperature, minimum temperature, rainfall and solar radiation. Both maximum temperature and minimum temperature show the same trend, but the model slightly underestimates the maximum and minimum temperatures. Rainfall plays a vital role in determining the growth and development of rice. It is seen that the annual cycle of rainfall is followed by the model, but it overestimates the rainfall in southwest monsoon season (June, July, August and September). Next variable on priority for rice growth is solar radiation, which has a close match with the observation.

In order to establish climate change scenario, it is important to establish the changes in climatological variables for present-day (1994–2003) (hereafter called as the base period) and for future RCP8.5 scenario (2090–2099) (hereafter called as the scenario). For the base period, we make use of observed data. For future RCP8.5 scenario, we use

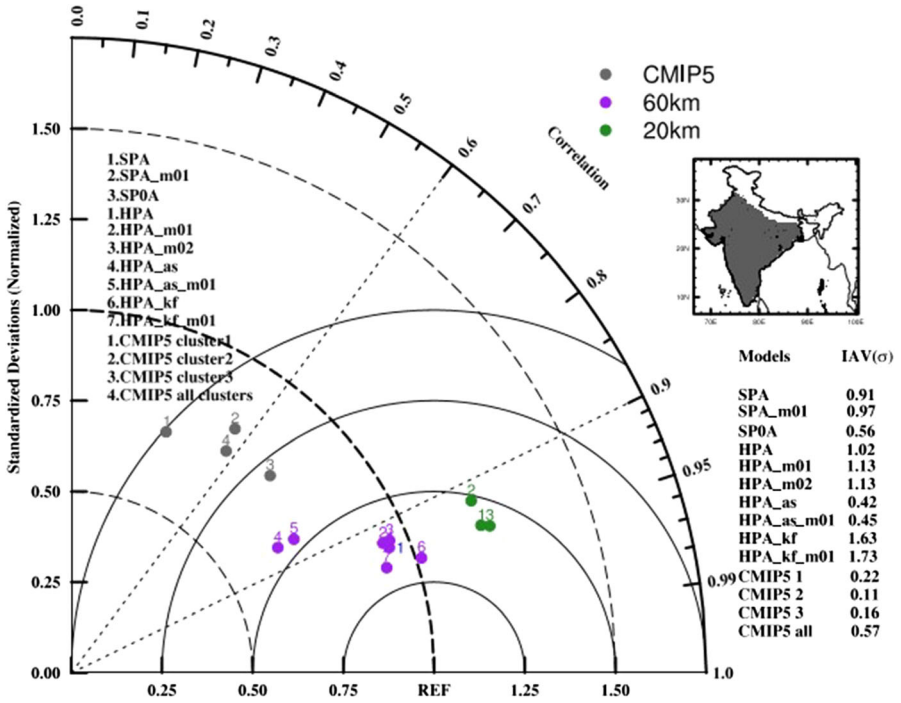


Figure 3. Taylor diagram comparing the skill scores (viz., normalized standard deviation, pattern correlation coefficient and bias) for 20 km MRI model against the current state of the art climate models, in simulating the present day Indian summer monsoon rainfall (after Varghese *et al.* 2020).

bias corrected model output. Model output needs to be bias corrected for any impact assessment studies and it is done by the simplest method called delta change approach. This method works on model response to climate change in order to modify observations. For variables like maximum temperature, minimum temperature and solar radiation, future climate variable can be derived as:

$$S_{\text{corr}} = \text{Obs} + \left(\bar{S}_{\text{fut}} - \bar{S}_{\text{pre}}\right), \qquad \text{(i)}$$

where S is the simulated variable with suffices 'corr', 'fut' and 'pre' referring to bias corrected, future and present simulations, respectively. 'Obs' stands for actual observed value. The overbar represents climatology corresponding to the period under consideration.

For rainfall, bias correction is performed as:

$$S_{\text{corr}} = \text{Obs} * \frac{\bar{S}_{\text{fut}}}{\bar{S}_{\text{pre}}}. \qquad \text{(ii)}$$

The annual climatology of present-day observed and bias corrected future maximum temperature, minimum temperature, rainfall and solar radiation are given in figure 5.

CERES Rice module within DSSAT package is a process oriented management level model of rice (*Oryza sativa* L.) for crop growth and development by considering the effects of weather, soil water balance, carbon and nitrogen balance, management and crop genetics. The components within DSSAT are the driver program (simulation timing control), land unit module (for managing all processes which affect land), primary modules (to separately simulate processes that affect land, viz., weather, plant growth, soil processes and their association with each other along with crop management practices). Many versions of it have been in use ever since its initiation in 1985 (Ritchie 1986; Godwin *et al.* 1990). The Cropping System Model (CSM) CERES has been used to simulate rice yield under various crop management practices and climate change scenarios. Also, it has been widely used all over the world (Rosenzweig and Parry 1994; Lal *et al.* 1998; Aggarwal and Mall 2002) for investigating climate change impact on rice yield. Thus, there is wide opportunity to use this as a tool for determining crop management for increased rice productivity. For running DSSAT CSM, input data required are weather data, soil data and crop management data. We used DSSAT version 4.6 for the purpose of yield estimation.

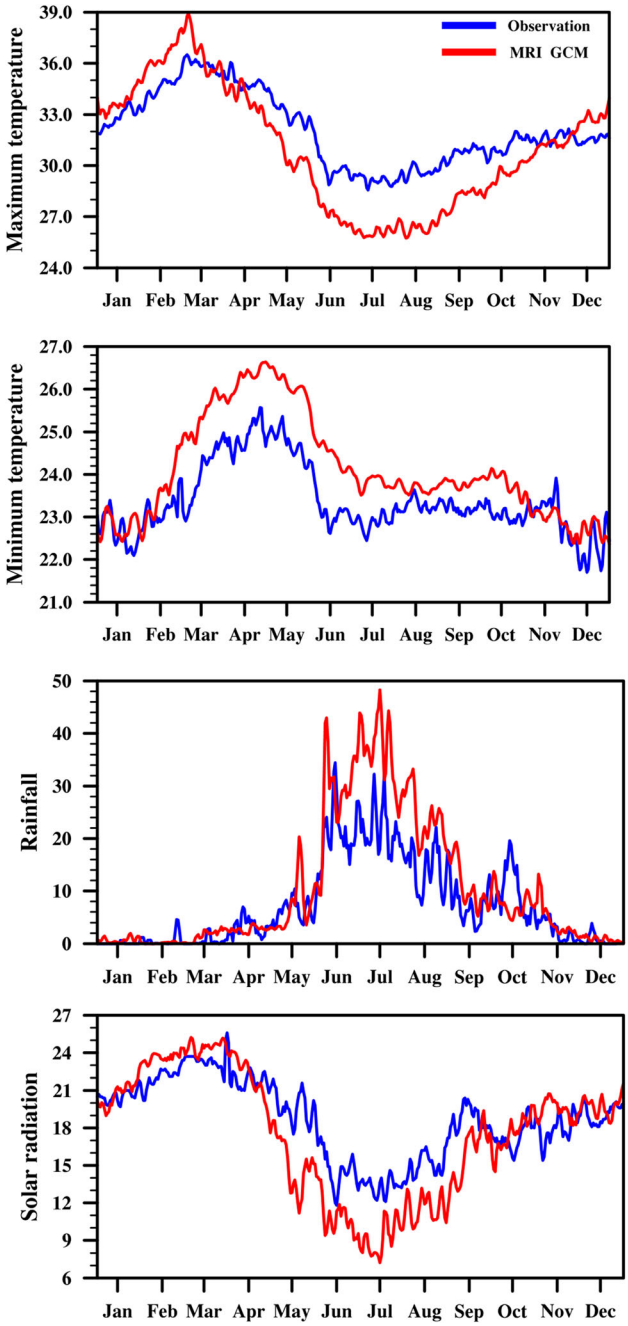


Figure 4. Observed and simulated climate parameters for the base period over Vellanikkara. Climatological mean (1994–2003) annual variation of minimum and maximum temperatures (°C), rainfall (mm/day) and solar radiation (MJm^{-2}/day) for the base period.

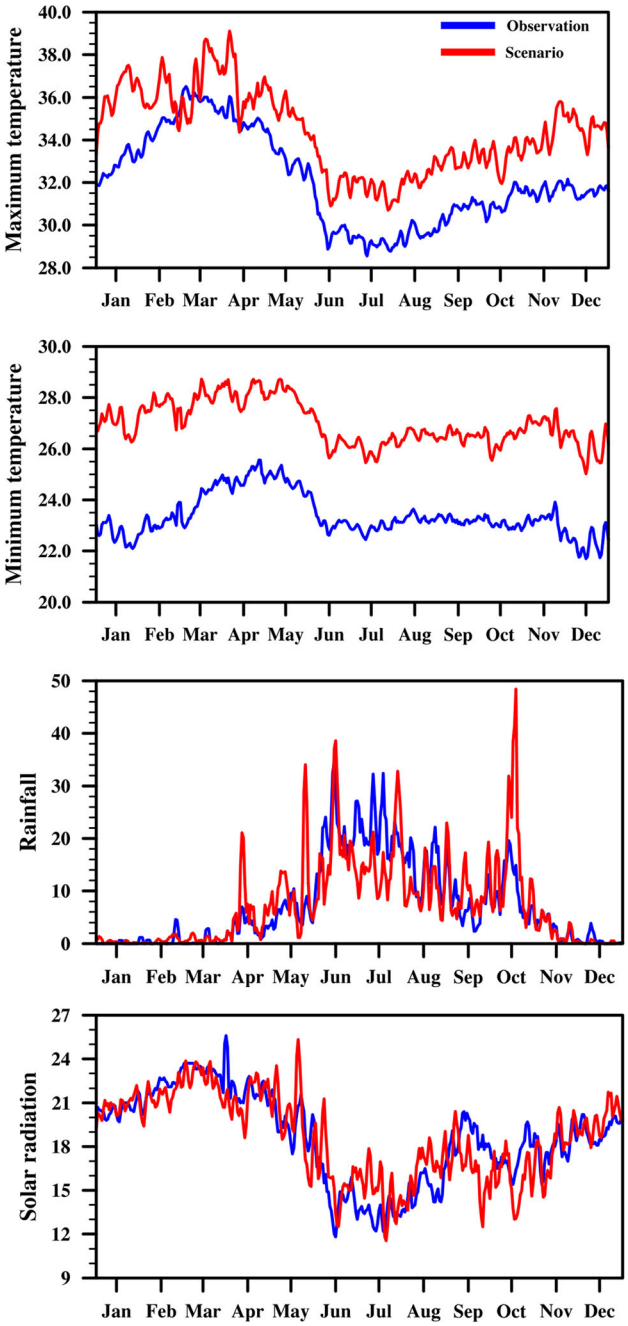


Figure 5. Climate parameters observed for the base period (1994–2003) and projected for future RCP8.5 scenario (2090–2099) (bias corrected) over Vellanikkara. Climatological mean of annual variation of minimum and maximum temperatures (°C), rainfall (mm/day) and solar radiation (MJm^{-2}/day) for the base period and for future scenario.

2.7 *Model calibration and testing*

Crop specific genetic coefficient data needs to be given as input into DSSAT. Thus, best cultivar coefficient is derived prior through iterative process of calibration based on crop type and location. This allows for performance of model for different rice varieties. Model calibration is the adjustment of parameters for comparing simulated values with observed values. The genetic coefficients were derived through iteration process by adjusting the relevant coefficients so as to attain maximum match between the simulated and observed days taken for each phenological events. This will make use of crop performance data such as planting date, plant density, row spacing, fertilizer application, irrigation, date of

panicle initiation, date of physiological maturity, date of harvesting, harvesting methods, grain yield and leaf area at different growth stages. The crop weather relationship in rice variety *Jyothi* was studied by Vysakh *et al.* (2016) to validate the CERES Rice model and to calibrate the genetic coefficients of rice at the Department of Agricultural Meteorology, Kerala Agricultural University, Thrissur. The field experiments were carried out during the *Kharif* seasons of 2013 and 2014 and are similar to the present study. Genetic coefficients were calibrated through intensive iterative process which involved over 6000 iterations and the model was validated. With the calibrated values, the model was found to simulate the phenology of *Jyothi* with a Root Mean Square Error (RMSE) of 0.89 for physiological maturity. Table 4 shows the calibrated genetic coefficients for *Jyothi* as in Vysakh *et al.* (2016) which is used in the present study.

A model is first required to be validated to provide guidance on practical management in agricultural system. Since the final yield of rice is regarded as the most important variable to be considered, the experimental plot yield is compared to the simulated yield. The CERES Rice model within DSSAT was validated for the short duration variety of *Jyothi* during wet season of 2017. Table 5 shows the comparison of yield obtained and simulated yield for 2017. Change in yield between simulated and observed can be given by percentage change in yield as:

$$\text{Percentage change in yield}\,(\Delta Y) = \frac{Y_s - Y_o}{Y_o} \times 100, \quad \text{(iii)}$$

where Y_s is yield simulated using CERES model [in kg/ha] and Y_o is yield observed [kg/ha] at the experimental site for the year 2017. It can be seen that model is capable of simulating the yield close to the observed value. The RMSE of the quantities in table 5 is found to be 660 kg/ha, suggesting a moderate error in the simulated yield.

3. Results

3.1 *Simulation for base period and scenario*

After calibration, simulations were carried out for the base period and for future scenario to calculate the respective crop growth parameters. Simulations were done to calculate the potential yield from which the model itself can take care of the nutrient and water requirements. Only climate parameters are taken into the model as user input. The CO_2 concentration is kept at a constant value of 380 ppm. Although the concentration of CO_2 is going to vary in future climate and can interfere in crop growth and development, we restricted our study to a constant value (380 ppm) and focussed only on the effects of climatological variables. An analysis of minimum temperature, maximum temperature, rainfall and solar radiation for base period and scenario will indicate how different the climatic variables are going to behave. Table 6 shows the comparison for the same. We can see a consistent increase in maximum and minimum temperatures, whereas a decrease in rainfall especially in the initial dates of planting for the station under consideration.

Fu *et al.* (2008) found from control experiments that the major effect of high temperature (daily

Table 4. *Calibrated genetic coefficient for rice variety Jyothi, after Vysakh et al. (2016).*

Genetic coefficients	Description	Calibrated genetic coefficients for *Jyothi*
P1	Basic vegetative phase, the time period expressed as growing degree days (GDD) in °C above a base temperature of 9°C	553.0
P2R	Photoperiod sensitivity coefficient, extent to which the phase development leading to panicle initiation is delayed	22.3
P5	Time period in GDD °C from beginning of grain filling to physiological maturity with a base temperature of 9°C	444.0
P20	Critical photoperiod or the longest day length in hours at which the development occurs at a maximum rate	10.0
G1	Potential spikelet number per panicle	51.0
G2	Single grain weight (g)	0.0250
G3	Tillering coefficient	1.10
G4	Temperature tolerance coefficient	1.10
PHINT	Phyllocron interval	82.0

Table 5. *Observed and predicted yield. Percentage change in yield is shown in 4th column.*

Planting dates	Yield observed for 2017 (kg/ha)	Yield simulated for 2017 (kg/ha)	ΔY (%)
D1	5750	6129	6.5
D2	6320	6234	−1.3
D3	5317	6338	19.2
D4	6430	6541	1.7
D5	5435	6422	18.1

Table 6. *Climate parameters, viz., maximum temperature (°C), minimum temperature (°C), rainfall (mm) and solar radiation (MJm^{-2}/day) for various dates of planting (given as D1–D5).*

Date of planting	Climate parameters	Base period	RCP8.5 scenario
D1	T_{max}	29.6	32.0
	T_{min}	23.1	26.3
	Rainfall	1800	1364
	S_{rad}	14.9	15.8
D2	T_{max}	29.8	32.2
	T_{min}	23.1	26.3
	Rainfall	1554	1189
	S_{rad}	15.5	16.1
D3	T_{max}	29.9	32.4
	T_{min}	23.1	26.4
	Rainfall	1390	1101
	S_{rad}	15.9	16.0
D4	T_{max}	30.2	32.5
	T_{min}	23.2	26.4
	Rainfall	1173	1241
	S_{rad}	16.6	16.1
D5	T_{max}	30.6	32.9
	T_{min}	23.2	26.5
	Rainfall	950	1086
	S_{rad}	17.2	16.5

average temperature and maximum temperature) on rice is shortening of grain filling phase and thus reduced seed setting rate, which in turn came from increased spikelet sterility. Thus, maintaining high spikelet fertility is necessary to prevent loss in yield. Wahid *et al.* (2007) analysed the heat stress due to high temperature on rice growth and development and ultimately its yield. It was found that temperature fluctuations can adversely affect tillering, resulting in less productive tillers and poor seed setting in rice crop. The effect of agrometeorological parameters such as maximum and minimum temperature, solar radiation, total rainfall, relative humidity and wind speed on rabi rice in Tamil Nadu were analysed by Kaladevi *et al.* (2000). They found a high dry matter production due to increase in minimum temperature. The effect of temperature on duration of tillering was studied by Lalitha *et al.* (2000) in lowland rice varieties grown in Andhra Pradesh. They reported that temperature during tillering period played a decisive role in determining the duration of the crop, with duration increasing with decreasing temperature and vice versa. The effect of quantum of rainfall, number of rainy days and length of rainy season on the productivity of rice in Jabalpur was studied by Gupta *et al.* (2000). They found correlation between such variables and the rice yield and their study concluded that, for obtaining high productivity, the moisture stress during reproductive phase and grain filling phase should be curbed by efficiently using the surplus rainfall.

Over the past century, studies have indicated that with increased atmospheric greenhouse gas concentration, the daily minimum temperature (night-time temperature) has increased more than daily maximum temperature (Easterling *et al.* 1997). In a modelling study by Varghese *et al.* (2020), similar result was obtained for future climate projections with increased CO_2 concentrations over India. Hence, a warming world has more influence on daily minimum temperature. An increase in minimum temperature was found to reduce the rice yield (Peng *et al.* 2004; Wassmann *et al.* 2009). Auffhammer *et al.* (2012) found that the night-time warming induced by global warming had greater impact on rice yield than any other changes in monsoon characteristics. Ramdoss and Subramaniam (1980) reported that rice yield has been positively affected by maximum temperature and number of sunshine hours and affected negatively the straw yield. Similarly, reduced solar radiation also decreases rice yield (Praba *et al.* 2004). The effect of climate change and monsoon in India was extensively studied by Auffhammer *et al.* (2012) which reported negative effect of drought and extreme rainfall on rice yield during *Kharif* season. Sensitivity experiments were done on rice by Bhattacharya and Panda (2013), which reported decrease in rice yield due to increase in temperature and an increase in yield due to increase in rainfall. Another temperature sensitivity experiment was done by Rani and Maragatham (2013), which found that increased temperatures during growth period has significantly affected the rice phenology and reduced the days to attain maturity, resulting in decreased yield.

3.2 *Growth and development variables using simulation*

The potential yield by the crop model is expected to have the effect of climatic parameters alone by taking into account all other crop management parameters by itself. Table 7 shows how the model has simulated the phenophase and yield output of rice with two different climate scenarios.

It is observed that the panicle initiation day and anthesis day has shifted to an early period (a decrease of ∼4 days and ∼7 days) in future scenario. The physiological maturity date has also undergone shift to an early maturity in the scenario case (∼8 days). The leaf area index is also decreasing as we go from base period to scenario. This would mean a decrease in total biomass production. It is also seen from table 7 that the potential yield (yield at harvest maturity) is decreasing in the future scenario (a decrease of ∼957 kg/ha). A delay in planting date increases the yield, especially when the date of planting corresponds to D4 which is 20th July, which is observed in both base period and scenario.

Under RCP8.5 climate scenario, the grain yield of rice variety *Jyothi* declined by 12.5%–17.6% at the end of the 21st century. The environmental factors such as temperature, rainfall and solar radiation of various dates of planting and at various growth phases are analysed to see how in future the date of planting is going to affect the simulation yield. Table 8 shows environment factors for various dates of planting as was simulated by the model. The average temperature in the rice growing season during the base period was 26.4°–26.9°C and increased to 29.2°–29.7°C during scenario period, thus indicating an average rise in temperature of 2.8°C. The results of simulation show that there is an increase of 1.7°–2.7°C in maximum temperature and 2.8°–3.6°C in minimum temperature. Increase in minimum temperature during the flowering stage will increase the respiration losses and lead to reduction in the yield (Peng *et al.* 2004) and increase in maximum temperature will shorten the length of the grain filling phase and reduce the seed setting rate in rice (Fu *et al.* 2008).

An exposure to high temperature during flowering period can cause significant reduction in the seed setting rate and reduction in crop yield (Jagadish *et al.* 2008). Our result confirms the same in future climate scenario where temperatures go as high as ∼34°C. There is spikelet sterility induced by heat stress during grain filling phase which reduces the grain filling capacity and the yield (Fitzgerald and Resurreccion 2009). Our results are also in confirmation with Krishnan *et al.* (2007), which reported that high maximum temperature during flowering phase significantly reduced the yield of rice. These have resulted in the decrease in number of days to panicle initiation, days to anthesis and days to physiological maturity as we go from base period to future RCP scenario.

The duration of sunshine hours or the solar radiation is found to increase from D1 to D5 which is yet another reason for increase in rice yield as we go from D1 to D5. For the base period, we find an increase in maximum temperature as we go from D1 to D5 along with a reduction in rainfall. This has resulted in reduced number of days for physiological maturity as we go from D1 to D5 (102–99 days). An increased maximum temperature showed a negative influence on the yield in delayed plantings, whereas rainfall during early planting periods favoured the high biomass. These results are supported by the findings of Kaladevi *et al.* (2000).

In the future scenario simulation, there is reduction in the rainfall compared to base period. Also, in the scenario, the rainfall is decreasing as

Table 7. *Growth and development variables. Base period and scenario comparison of growth and development variables using simulation for various dates of planting (given as D1–D5).*

	Base period					RCP8.5 scenario				
Growth variables	D1	D2	D3	D4	D5	D1	D2	D3	D4	D5
Panicle initiation day	35	36	36	35	34	31	31	31	31	30
Anthesis day	69	70	69	68	67	62	61	62	62	61
Physiological maturity day	102	102	101	100	99	92	92	92	92	91
Yield at harvest maturity	6236	6371	6584	6881	6831	5418	5447	5757	5871	5624
Number at maturity	24943	25485	26335	27525	27326	21672	21786	23028	23484	22497
Leaf area index	6.00	6.16	6.79	6.65	6.83	5.03	5.04	5.30	5.59	5.30
Harvest index at maturity	0.411	0.402	0.408	0.420	0.414	0.414	0.408	0.426	0.428	0.424

Table 8. *Environment factors for base period and future climate scenario.*

Dates of planting		Base period						RCP8.5 scenario					
		1	2	3	4	5	6	1	2	3	4	5	6
D1	A	23	29.9	23.1	26.5	14.5	495.1	20	32.2	26.4	29.3	15.9	395.0
	B	12	29.4	22.9	26.1	13.7	263.2	11	32.0	26.2	29.1	15.8	157.2
	C	34	29.2	23.0	26.1	13.7	659.0	31	31.6	26.1	28.8	14.7	481.9
	D	7	29.8	23.4	26.6	16.0	90.8	6	32.3	26.6	29.4	15.8	59.3
	E	25	30.0	23.3	26.6	17.0	286.1	23	32.4	26.6	29.5	17.2	100.2
	F	102	29.6	23.1	26.4	14.9	1800	92	32.0	26.3	29.2	15.8	1364
D2	A	23	29.5	22.9	26.2	13.9	499.4	20	32.0	26.2	29.1	15.9	298.0
	B	13	29.2	22.9	26.0	13.3	277.8	11	31.6	25.7	28.7	14.7	155.5
	C	34	29.4	23.2	26.3	14.6	522.8	30	31.7	26.4	29.1	15.4	419.8
	D	6	29.9	23.3	26.6	17.3	76.5	7	32.3	26.6	29.5	16.9	82.4
	E	25	30.8	23.3	27.0	18.9	164.1	23	33.0	26.5	29.8	17.6	232.4
	F	102	29.8	23.1	26.4	15.5	1554	92	32.2	26.3	29.2	16.1	1189
D3	A	24	29.1	22.9	26.0	13.3	526.3	20	31.8	25.9	28.9	15.1	264.8
	B	12	29.2	23.1	26.2	14.0	202.5	11	31.1	26.3	28.7	13.9	214.7
	C	33	29.9	23.3	26.6	16.4	392.0	31	32.4	26.5	29.5	16.8	342.3
	D	7	30.8	23.1	27.0	19.8	44.4	6	33.3	26.7	30.0	18.9	40.6
	E	24	30.9	23.2	27.0	17.7	205.3	23	33.2	26.5	29.8	16.0	228.3
	F	101	29.9	23.1	26.5	15.9	1390	92	32.4	26.4	29.4	16.0	1101
D4	A	23	29.2	23.1	26.2	13.9	398.2	20	31.5	26.3	28.9	14.2	310.6
	B	12	29.7	23.4	26.6	15.5	173.5	11	32.2	26.6	29.4	17.2	114.9
	C	33	30.4	23.2	26.8	18.0	277.1	31	32.8	26.5	29.7	17.4	315.9
	D	7	31.1	23.3	27.2	17.8	64.7	6	33.0	26.5	29.8	15.0	51.7
	E	24	31.0	23.1	27.0	17.1	255.7	23	33.1	26.3	29.7	15.8	413.4
	F	100	30.2	23.2	26.7	16.6	1173	92	32.5	26.4	29.5	16.1	1241
D5	A	23	29.6	23.3	26.4	15.1	330.0	20	32.1	26.6	29.4	16.8	222.3
	B	11	30.0	23.3	26.6	17.3	121.7	10	32.8	26.5	29.6	17.0	115.2
	C	33	30.8	23.2	27.0	18.4	241.5	31	33.2	26.5	29.9	16.6	283.8
	D	6	30.8	23.0	26.9	16.2	99.2	6	33.3	26.0	29.6	16.5	68.9
	E	25	31.5	23.1	27.3	17.7	155.3	23	33.2	26.5	29.8	15.8	393.5
	F	99	30.6	23.2	26.9	17.2	950.7	91	32.9	26.5	29.7	16.5	1086

A: Emergence to end juvenile, B: End juvenile to panicle initiation, C: Panicle initiation to end of leaf growth, D: End of leaf growth to beginning of grain filling, E: Grain filling phase, F: Planting to harvest.

1: Time span in days, 2: Maximum temperature (°C), 3: Minimum temperature (°C), 4: Mean temperature (°C), 5: Solar radiation (MJm^{-2}/day), 6: Rainfall (mm).

we go from D1 to D5. Subsequently, the yield in future increases from D1 to D5. The rainfall at different growth stages determine whether it is beneficial or not for the final yield of rice. Decrease in rainfall during reproductive stage results in better yield as reported by Ghildyal and Jana (1967). From the study we obtained similar results as we go from D1 to D5 which may be the reason for increased yield at later planting dates. If delayed planting date is giving more yield in future in the central zone of Kerala, there is a need to shift the date of planting to last week of July or first week of August.

Moreover, in the climate change scenario, the effect of CO_2 is not clearly understood since it is kept at a constant value of 380 ppm. If the direct physiological effect due to increased CO_2 were also taken into consideration, a more robust result could be obtained, although it may not separate the impact on changes in climate parameters.

Leaf Area Index (LAI):

LAI was suggested by Williams (1946).

$$\text{Leaf area index} = \frac{\text{Total leaf area of plant}}{\text{Land area occupied by plant}}. \quad \text{(iv)}$$

It is usually calculated at 15-day interval from transplanting to harvesting using LAI from randomly selected plants. This index can be

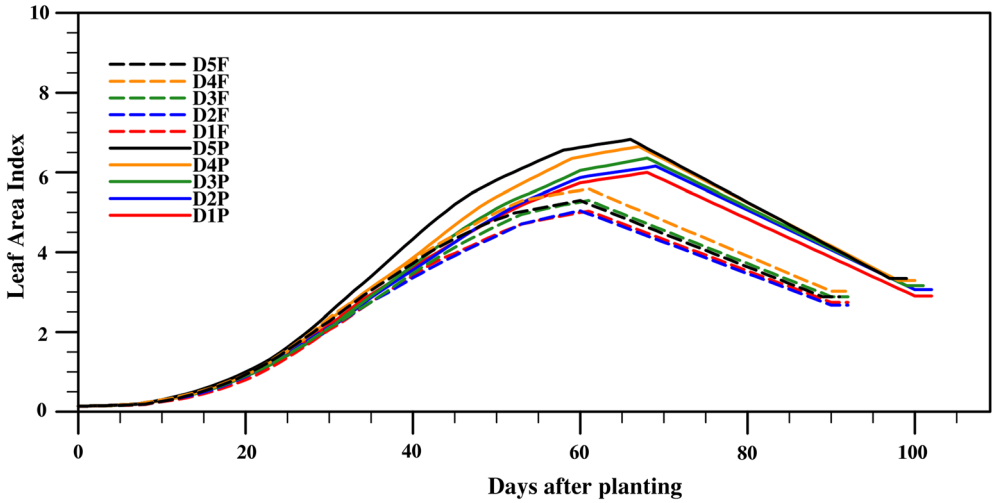


Figure 6. Leaf area index (LAI) for rice crop for base period (solid lines) and future scenario (dashed lines). Different days of planting are given as D1–D5 with P representing base period and F representing future scenario.

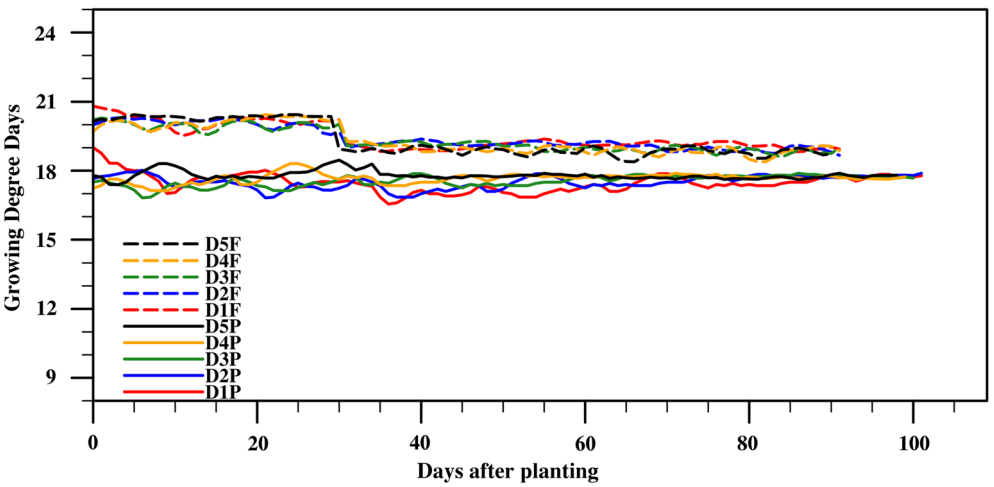


Figure 7. Growing degree days (GDD) for rice crop for base period (solid lines) and future scenario (dashed lines). Different days of planting are given as D1–D5 with P representing base period and F representing future scenario.

used as a proxy to derive the total biomass yield as well as yield at maturity and hence, it is interesting to note the evolution of LAI in rice crop. It can be seen from table 7 that the LAI decreases as we go from base period to future scenario. Figure 6 shows the same for base period and future scenario. It is evident that the base period has more leaf area index compared to that of scenario, suggesting a better biomass yield and yield at maturity in the present. Maximum LAI is found during 60 days after transplanting. The decrease in LAI towards maturity is because of decreased leaf number which is due to senescence of leaves that were formed. Positive correlation exists between LAI and harvest yield of different dates of planting in both base period and future climate scenario.

3.2.1 *Grain number*

This is a biometric observation technique followed to quantify yield. It is a measure of total number of grains at harvest maturity. By analysing this variable, we can get an estimate of the yield for base period and future scenario. From table 7, it can be observed that the grain number decreases as we go from base period to scenario, implying a decrease in the overall yield ($\sim$3829) in the future.

3.2.2 *Growing degree days (GDD)*

It is a quantity which is calculated for the entire growth period and is used to relate the effect of temperature on the growth duration and crop yield. It is calculated using the following equation.

The base or threshold temperature used in the calculation of GDD is assumed as 10°C for rice (Islam and Sikder 2011).

$$\mathrm{GDD} = \sum_{i=0}^{n} \frac{T_{\max} + T_{\min}}{2} - T_b, \qquad \text{(v)}$$

where n is the number of days from sowing date till the date of harvesting, $T_{\max}$ is the maximum temperature (°C), $T_{\min}$ is the minimum temperature (°C), and T_b is the base temperature (minimum threshold temperature).

Figure 7 shows the GDD for base period and scenario runs. It can be seen that the GDD for base period is less compared to scenario case. This would mean that the crops in base period have survived for longer duration at a much lesser temperature compared to the scenario case. The development of crops depends on accumulation of heat and if the heat is accumulated at once, the growth duration ceases and the yield decreases.

4. Conclusions

The major focus of this study was to determine the importance of climate variables in the prediction of yield for present and future climate scenarios. The purpose of crop models is to predict the crop yield under the prescribed management characteristics wherein actual production are usually hindered by factors outside the purview of model. This would result in the model overestimating the yield and hence the gap between simulated and actual yields. Yield prediction was done using CERES Rice CSM within DSSAT for model validation using actual field observations and CSM was found to match with only slight overestimation in the yield. Weather parameters such as maximum temperature, minimum temperature, rainfall and solar radiation were analysed for the station for base period and bias-corrected for future RCP8.5 scenario. A clear increase in maximum and minimum temperature was seen whereas rainfall was found to decrease.

Further, crop simulations were done for different dates of planting for base period and RCP8.5 scenarios to find the change in potential yield. It was found that the increase in temperature affects the crop duration which indirectly affects the yield over the station under consideration. The growth and development variables such as panicle initiation day, anthesis day, physiological maturity day, yield at harvest maturity, number at maturity, leaf area index and harvest index at maturity were found to decrease as we go from base period to scenario. Thermal time requirement is one of the key factors responsible in a crop which determines its yield. GDD was thus found to decrease during the scenario shift, indicating early maturity of rice crop in future scenario. Another factor which accounts for it would be tolerance to heat stress.

In this study, we have only analysed the predicted value of rice yield for climate change scenario at constant CO_2 and varying meteorological variables and found a decrease in yield for all planting dates. The effect of CO_2 needs to be introduced so as to identify combined effect of increase in CO_2, temperature and solar radiation on different phenophases of rice. From our findings, we suggest for shift in planting date of *Kharif* crop to last week of July or first week of August for the central zone of Kerala in future. We suggest for introduction of temperature tolerant and high yielding varieties using genetic approaches (Ogden and Innes 2008). Selection of short duration varieties that are drought tolerant may also be suggested for better water use efficiency.

The current study is only a pilot study to evaluate the feasibility of using very high resolution ($\sim$20 km) GCM for climate change impact studies on rice crop which is different from the existing studies which make use of downscaled coarse resolution GCM data. Currently, this study is limited to a single grid owing to unavailability of station-based weather, crop, and soil data. We intend to extend this work in future to the rainfall homogeneous zones of India, in order to establish the usefulness of this GCM.

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